

- 1 (a) State the *principle of moments*.

..... [1]

- (b) It is said that Archimedes used huge levers to sink Roman ships invading the city of Syracuse. A possible system is shown in Fig. 1.1 where a rope is hooked on to the front of the ship and the lever is pulled by several men.

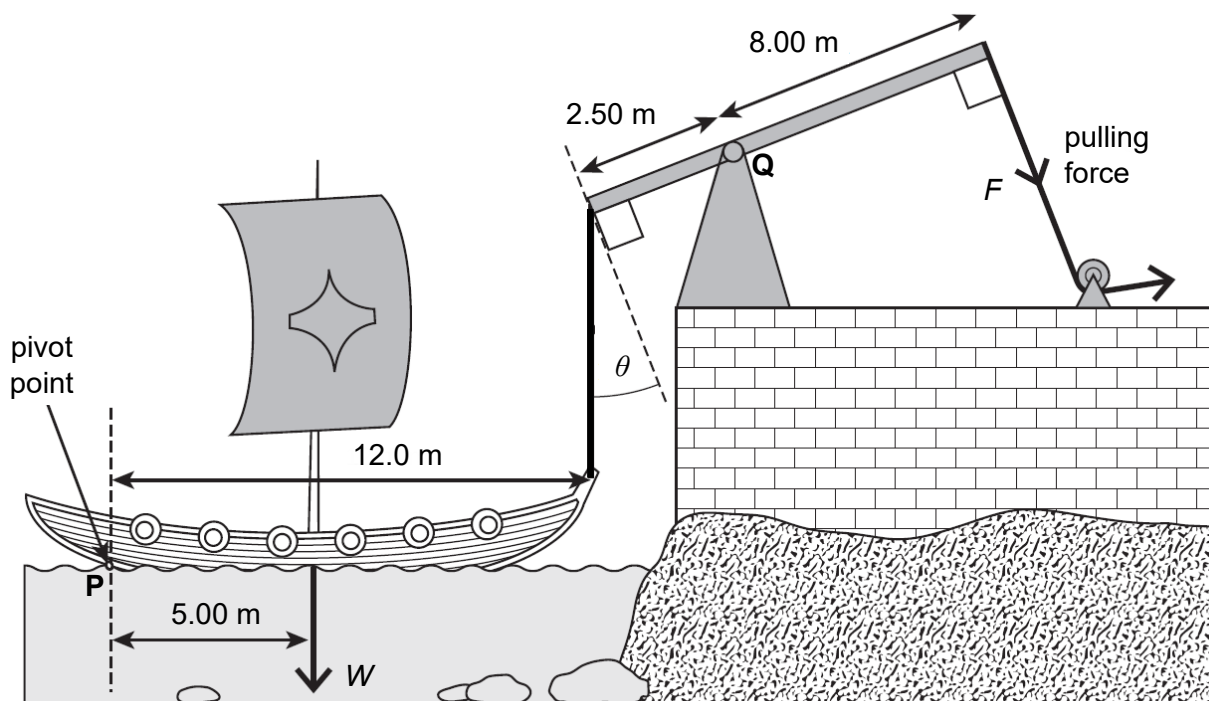


Fig. 1.1

When the ship is just lifted from the water, upthrust is negligible.

The weight W of a Roman ship was 3.40×10^4 N.

The rope hooked to the front of the ship is vertical. The angle θ is 24.0°

The centre of gravity of the light beam of the lever is at point Q.

- (i) Assume the ship pivots about point P. Show that

1. the tension in the rope hooked to the front of the ship is 14.2 kN,

[1]

2. the force F is 4.05 kN.

[2]

- (ii) A force R is exerted on the beam of the lever at point Q at an angle α to the vertical, as shown in Fig. 1.2.

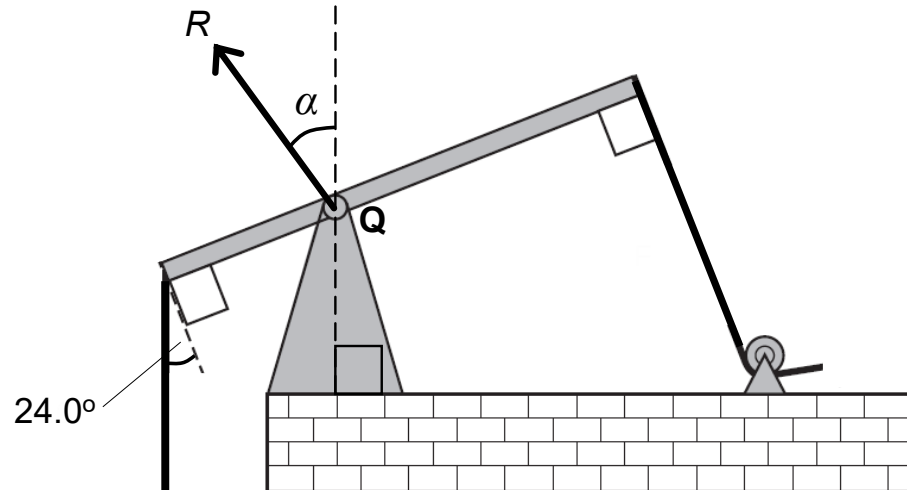


Fig. 1.2

Use your answer in (i) to determine the magnitude of R and the angle α .

magnitude of force = N

angle α = $^\circ$

[3]

- (c) Fig. 1.2 shows a trailer attached to the towbar of a stationary car.

All forces acting on the stationary trailer are vertical. The weight W of the trailer acts through its centre of gravity. F is the force exerted by the towbar on the trailer. F_R is the **total** normal reaction force experienced by the trailer.

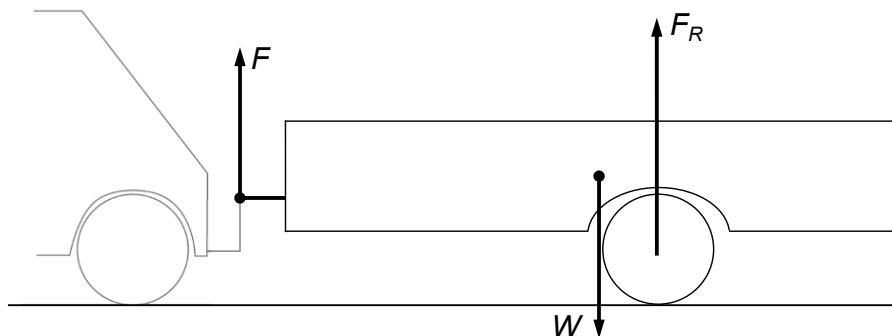


Fig. 1.2

State and explain the changes to the magnitude and direction of the force F when the car starts to move forwards.

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..... [3]